

Aritra Mondal

Data Scientist | Machine Learning & Distributed Systems

Production ML (Amadeus) • Applied MSc Bac+5 (Summa Cum Laude / Rank 1) • Immediate Availability (CDI Cadre)

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Location: Paris & Biarritz, France (Relocation Ready / Hybrid) • **Work Rights:** Titre de séjour (Full EU Rights, No Sponsorship) • **Notice:** Immediate (0-Day) – CDI Cadre

PROFESSIONAL SUMMARY

Production-focused **Data Scientist** with a proven enterprise track record at **Amadeus** scaling distributed machine learning and big data pipelines for airline revenue management and demand forecasting. Specialized in **time-series predictive modeling**, high-throughput distributed computing (**Apache Spark, Scala, AWS EMR**), and automated **MLOps CI/CD**. Combines academic excellence from an **Applied MSc in Data Science & AI (Summa Cum Laude, Rank 1 / Félicitations du Jury, Bac+5)** with deep systems fundamentals from a **BSc (Hons) Computer Security (First Class)**. Authorized to work across France & EU with zero notice period.

TECHNICAL COMPETENCIES

Machine Learning & AI:	**Python**, **Scikit-learn**, **XGBoost**, LightGBM, **Time-Series Forecasting**, PyTorch, LSTM, Feature Engineering, Bayesian Optimization (**Optuna**), TreeSHAP explainability.
Big Data & Engineering:	**Apache Spark**, **Scala**, **PySpark**, **AWS EMR**, **Parquet**, Amazon S3 Data Lakes, Advanced SQL (Window Functions, CTEs, Joins), DuckDB, Neo4j.
Cloud, MLOps & Systems:	**AWS (EMR, S3, EC2, IAM)**, **Docker**, **Jenkins CI/CD**, Linux/UNIX Internals, Bash Shell Scripting, Git Workflows, Asynchronous REST APIs.
Cybersecurity & Protocols:	**TCP/IP, UDP, ICMP Telemetry**, **SNORT Network IDS**, Deep Packet Inspection, Heuristic Rule Tuning, Cryptography, Defensive System Architecture.

PROFESSIONAL EXPERIENCE

Data Scientist (Apprentice) — Amadeus
Aug 2023 – Sep 2024

Sophia Antipolis & Remote, France · Airline Revenue Management & Demand Forecasting (Order Business Domain)

- **Production ML Architecture:** Engineered and deployed scalable end-to-end machine learning pipelines for passenger demand forecasting and revenue optimization across distributed **AWS EMR (Spark/Scala)** clusters and **Amazon S3** data lakes.
- **Terabyte-Scale Booking Analytics:** Ingested and transformed complex enterprise booking databases containing **300+ attributes and millions of PNR-linked records** (origin, destination, layovers, booking windows) with historical coverage extending to 2010.
- **Catalyst Optimization & Migration:** Accelerated distributed batch ETL processing throughput by **35%** by migrating legacy PySpark workflows to native compiled **Scala**, optimizing partition schemas, eliminating shuffle spills, and tuning Catalyst plans.
- **Automated MLOps Pipelines:** Built robust automated **CI/CD workflows** using **Jenkins and Bash**, standardizing reproducible test suites, artifact deployments, and expected-vs-actual verification, cutting manual intervention by **50%**.
- **Data Integrity & Model Ingestion:** Conducted rigorous functional validation and anomaly detection on partitioned **Parquet** data lakes, ensuring high-fidelity feature feeds for downstream high-throughput predictive models.

Android / Software Developer (Intern) — Jungleworks
Nov 2020 – Apr 2021

100% Remote · Mobile Software Architecture & Distributed REST API Integrations

- **Native Client Engineering:** Developed native Android client software in **Java / Kotlin**, integrating asynchronous **REST APIs** with defensive programming and robust network error recovery protocols.
- **Offline-First Persistence:** Architected an **offline-first local caching architecture (SQLite)**, maintaining state consistency, preventing data loss, and mitigating transient network dropouts.
- **Quality Assurance & Stability:** Implemented structured **JUnit testing suites** and reproducible debugging workflows that eliminated core crash vectors and elevated production client stability.

FEATURED PRODUCTION & APPLIED PROJECTS

Airline Demand Forecasting Engine [Amadeus Production]
Distributed time-series forecasting system on **AWS EMR (Spark/Scala)**. Re-engineered Catalyst physical plans and partition schemes, delivering a verified **35% reduction in ETL execution latency** over multi-GB Parquet datasets.

High-Dimensional RecSys & Ranking Engine [Applied ML]
Multi-stage recommendation engine on 10,000+ items. Optimized ranking accuracy via Bayesian search (**Optuna**) and mitigated cold-start drop-off using **TreeSHAP feature explainability**.

Financial Time-Series & Alpha Engine [Open Source / Quant]
Automated data pipeline integrating **CCXT, yfinance, MT5** with **73+ engineered alpha indicators**. Evaluated XGBoost and LSTMs under **purged walk-forward cross-validation (0 lookahead leakage)**.

SNORT Network IDS Telemetry [Cybersecurity / Systems]
Real-time packet inspection for telemetry across **TCP/IP, UDP, ICMP**. Optimized heuristic inspection rules, reducing **false-positive alarms by ∼20%** without compromising detection coverage.

EDUCATION, CERTIFICATIONS & LANGUAGES

Education & Academic Honors

Applied MSc in Data Science & AI (Bac+5 Equiv.) 2022 – 2024
Data ScienceTech Institute (DSTI) · France · **Summa Cum Laude (Rank 1 / Major)**
Thesis: Spark/Scala ETL optimization & forecasting on AWS EMR. Alternance at Amadeus.

BSc (Hons) Computer Security (Bac+3 Equiv.) 2017 – 2020
Northumbria University · UK · **First Class Honours (Mention Très Bien)**
Thesis: Heuristic rule optimization for SNORT Network IDS (-20% false positives).

Advanced Diploma in IT (Bac+2 Equiv.) 2015 – 2017
MDIS Singapore · **Distinction in IT** · OOP (Java, C++), SQL, Distributed Systems.

Certifications & Verified Credentials

- **AWS Solutions Architect – Associate:** In Progress (2026)
- **Neo4j Certified Professional:** Neo4j Graph Academy (2024)
- **Certified Ethical Hacker (CEH):** EC-Council (ECC904251, 2019)
- **Fortinet NSE 1, 2, 3:** Certified (Valid through June 2026)

Languages

- **English:** Native / Bilingual (C2 – Professional & Technical)
- **French:** Professional Working Proficiency (B1/B2)
- **Hindi & Bengali:** Native / Bilingual